Ss 1A

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ make clean
make test_objects
./test_objects
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function ‘object_write’:
object.c:99:47: warning: ‘.tmp’ directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncation=]
   99 |     snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
      |                                               ^~~~
In file included from /usr/include/stdio.h:970,
                 from pes.h:8,
                 from object.c:3:
In function ‘snprintf’,
    inlined from ‘object_write’ at object.c:99:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:68:10: note: ‘__builtin___snprintf_chk’ output between 5 and 516 bytes into a destination of size 512
   68 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   69 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   70 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 1B

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 2A

```
find .pes/o
```

Then:

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization
```

SS 2B

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ xxd $(find .pes/objects -type f | head -1) | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 3A

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ ./pes init
echo "hello" > a.txt
./pes add a.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     a.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  commit.h
  untracked:  file.txt
  untracked:  tree.h
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  pes.c
  untracked:  commit.c
  untracked:  Makefile
  untracked:  object.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  README.md

vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 3B

```
./pes add a.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     a.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  commit.h
  untracked:  file.txt
  untracked:  tree.h
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  pes.c
  untracked:  commit.c
  untracked:  Makefile
  untracked:  object.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  README.md

vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776681777 6 a.txt
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 4A

SS 4B

SS 4C